Uncorrelated Sparse Autoencoder With Long Short-Term Memory for State-of-Charge Estimations in Lithium-Ion Battery Cells

Mayuresh Savargaonkar, *Member, IEEE*, Isaiah Oyewole, Abdallah Chehade, *Member, IEEE*, and Ala A. Hussein, *Senior Member, IEEE*

***Abstract*— For the safe and reliable operation of battery-driven machines, accurate state-of-charge (SOC) estimations are necessary. Unfortunately, existing methods often fail to identify patterns relevant to long-term SOC estimation due to complex battery cell characteristics such as aging. In this paper, we propose the Uncorrelated Sparse Autoencoder with Long Short-Term Memory (USAL). USAL is a novel neural network that addresses the challenging task of long-term SOC estimation given a limited initial history of a cell's charge-discharge behavior. USAL uses a multi-task learning strategy to harness the advantages of sparse autoencoders and Long Short-term Memory (LSTM) networks by enforcing correlation penalties. The USAL simultaneously learns to (i) generate a latent space of informative SOC encodings from commonly measured cell characteristics, (ii) penalize for high multicollinearity between encodings, and (iii) identify non-trivial long and short temporal correlations between the encodings using LSTM cells. USAL outperforms benchmarked models in our experiments when trained on five initial charge-discharge cycles across multiple battery cells using three publicly available accelerated aging datasets.**

***Note to Practitioners*—This paper proposes USAL, a custom-built deep neural network to address the challenging task of long-term SOC estimations in battery cells. Long-term SOC estimation involves estimating SOC for cycles near End-Of-Life (EOL) given some initial charge-discharge cycles. Three fundamental steps involved in long-term SOC estimations using USAL are (i) exploiting a multi-task learning strategy to learn efficient encodings given limited training data, (ii) penalizing these encodings for high correlations to efficiently transform measured inputs into a space of informative features, and (iii) mapping of aging-related trends to support long-term SOC estimations. USAL is designed to be a data-driven SOC estimation method that is (i) capable of alerting the user to a faulty cell when integrated into a real-life Battery Management System (BMS) and (ii) identifying the relative quality of a battery cell from only a few initial charge-discharge cycles.**

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***Index Terms*— Autoencoders, battery management systems, lithium-ion battery, machine learning, neural networks, state-of-charge.**

I. Introduction

LITHIUM-ION (Li-ion) battery cells have been pivotal in developing electric vehicles (EVs), drones, portable electronics, grid storage systems, and many other systems and applications. Today, these battery cells are used in various electronic devices due to the humongous spectrum of advantages they offer over existing energy storage technologies. Some advantages of Li-ion battery cells include their high specific energy, high cycle life, high Coulombic efficiency (up to 98%), and low self-discharge rate [1]. Their high efficiency and power density make the use of Li-ion battery cells extremely lucrative in modern energy-efficient electric grids [2]. Li-ion battery cells also offer significant performance improvements and other environmental benefits [3]. Thus, they are considered to be a sufficient alternative to existing energy storage systems and technologies.

Efficient management of Li-ion battery cells is necessary to ensure a safe operation, improved performance, optimized power management, and prolonged service life [4], [5]. These operations are commonly performed using a BMS. Figure 1 shows an overview of a BMS in an EV. For a BMS to work efficiently, accurately measuring the battery parameters is crucial. While battery measurements such as voltage, current, and temperature can be accurately measured, metrics such as SOC and state-of-health (SOH) can only be estimated within the control module using the measured inputs. SOC is a metric that is used to describe the ratio between the remaining capacity and the maximum (nominal) capacity of a battery measured in Ampere-hours (Ah) at a specific temperature [6]. Since Li-ion battery cells have non-linear characteristics that depend on various internal and external conditions, estimating the SOC is challenging [1], [7].

Xiong et al. [4] categorize SOC estimation methods into four distinct categories, namely, (i) look-up table methods, (ii) integral methods, (iii) model-based methods, and (iv) data-driven methods. Look-up table methods are commonly used to map measured characteristics such as open-circuit voltage (OCV) to SOC. This OCV-SOC relationship is pre-mapped, and the SOC estimations are performed by referencing and

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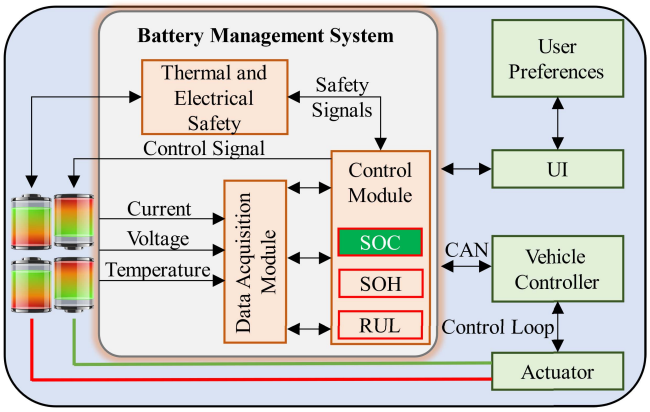


Fig. 1. High-level representation of a BMS in an electric vehicle.

locating the closest measured OCV value [8], [9]. However, while look-up table-based methods offer accurate SOC estimations, they cannot perform online estimations needed by EVs due to isolation requirements [1], [10], [11].

Integral methods such as Coulomb counting use integrals of cell current measured over time to calculate the SOC.

$$SOC_t = SOC_{t_0} + \frac{1}{C_N}\int_{t_0}^{t} \eta i dt \tag{1}$$

In eq. 1, SOC_t is the SOC estimated at time t and SOC_{t_0} is the reference SOC at some time t_0, C_N is nominal cell capacity, η is the cell efficiency, and i is the measured cell current in Amps. Although this method can perform online estimations, it is highly dependent on the reference value SOC_{t_0}. Further, it is also limited by its ability to filter noise in the measured current i. Thus, due to robustness issues caused by their open-loop nature, integral methods are complemented with model-based methods [4].

The estimation models are defined using highly complex and non-linear state equations in model-based methods. SOC estimation methods using Kalman filter-based approaches are a typical example of these methods [12]. Whereas model-based methods are more robust in their SOC estimations, they are often limited by their ability to capture cell aging and degradation information [13]. Although significant research has improved these models' accuracy, latency remains an unaddressed challenge [14]. Thus, researchers have come up with data-driven methods. These approaches are often referred to as 'black-box' methods due to their lack of explainability [4]. Although data-driven methods can be used for robust and online SOC estimations, they often use highly complex Machine Learning (ML) training procedures that require large amounts of training data for accurate SOC estimations [15], [16], [17], [18], [19]. Moreover, they often fail to account for cell aging and need frequent retraining [20].

Aging is a complex phenomenon experienced by Li-ion battery cells leading to capacity loss over time and use [21], [22]. Aging processes do not have one cause but result from several chemical processes and interactions [23]. Palacin [23] provides a detailed analysis of cell aging and its effects on capacity. Although cell aging is a well-known and researched phenomenon, little work has been done to adopt its effects on long-term SOC estimations. Short-term SOC estimation involves estimating the SOC for a few cycles in the immediate future. On the other hand, long-term SOC estimation involves estimating SOC for cycles near EOL, given some initial charge-discharge cycles. Researchers have recently tried to enhance the data-driven models using Transfer Learning (TL) with promising results for short-term SOC estimations [6], [24].

Although data-driven models show improvement upon other existing models, they often require hours of training on large amounts of battery aging data [6], [24]. Further, these models need frequent retraining to update their hyper-parameters. This helps them capture the latest aging-related temporal changes within the battery cell. Such models are also referred to as 'self-adaptive aging models.' A detailed review of such methods can be found in the work by Lucu et al. [25]. In similar work, researchers have also studied combining autoencoders and Long Short-Term Memory (LSTM) networks for short-term SOC estimations [26]. Although this study explores short-term SOC estimations in detail, it follows a two-step training procedure and fails to discuss the effect of correlated encodings on long-term SOC estimations.

This paper proposes the Uncorrelated Sparse Autoencoder with Long Short-Term Memory (USAL). USAL is a unique combination of data-driven sparse autoencoders and an LSTM network that performs accurate long-term SOC estimations by penalizing high correlations between encodings. Correlation penalties within USAL force it to transform commonly measured inputs (e.g., voltage) into a space of informative encodings that represent unique statistical features. The LSTM network within USAL then identifies temporal trends in this encoded data, supporting online and accurate long-term SOC estimations. In our experiments, USAL achieves accurate long-term SOC estimations when trained only on the initial five charge-discharge cycles under diverse operating conditions. USAL also achieves state-of-the-art when evaluated using dynamic loading profiles using a publicly available dataset. To the best of our knowledge, this is one of the first works that introduces the problem of long-term SOC estimations. This is also one of the first works that analyzes the effects of multicollinearity between encodings generated by an autoencoder for SOC estimations in Li-ion battery cells. In summary, some significant contributions of this work include:

1. **Long-Term SOC estimations:** Exploration of the challenges in long-term SOC estimations and proposition of a novel data-driven solution.
2. **Multi-task learning**: Proposition of a multi-task learning strategy to simultaneously train a sparse autoencoder and an LSTM network given limited training data.
3. **Informative encodings:** Learning a diverse and informative set of encodings by penalizing for high multicollinearity between learned encodings.
4. **Age-mapping:** Identifying and mapping aging-related trends in Li-ion battery cells for accurate and reliable SOC estimations.

The remainder of the paper is organized as follows; Section II introduces the USAL; Section III discusses the

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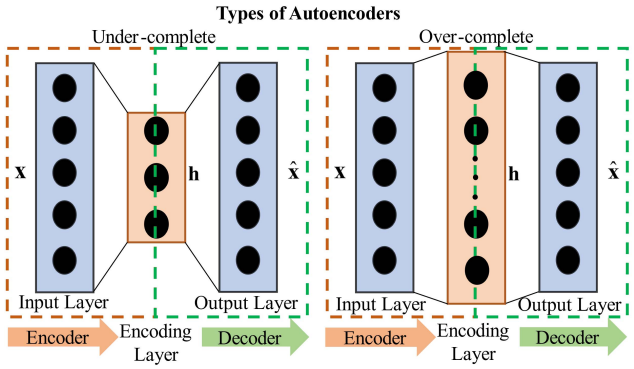


Fig. 2. Under-complete and over-complete autoencoders.

efficacy of the proposed model based on a case study; Section IV concludes the paper with directions for future work.

II. Uncorrelated Sparse Autoencoder With Long Short-Term Memory (USAL)

USAL is a state-of-art deep neural network that simultaneously learns to reconstruct inputs and perform long-term SOC estimations. The following discussion explores each part within the USAL in more detail.

A. Sparse Autoencoders

Autoencoders are specialized neural networks that are designed to transform inputs into a latent space of learned representations (encodings) [27], [28], [29]. Formally, encodings are defined as the 'learned representation of data that can reconstruct inputs even in the presence of noise' [27], [30], [31]. Figure 2 shows two autoencoders, each with one hidden layer. In this figure, an m-dimensional input vector $\mathbf{x} \in \mathbb{R}^m = X$ is mapped to a p-dimensional encoding $\mathbf{h} \in \mathbb{R}^p = F$ in the encoder ($\phi : X \rightarrow F$). In the decoder, the p-dimensional encodings are then transformed to an m-dimensional reconstruction $\hat{x} \in \mathbb{R}^m = \hat{X}(\psi : F \rightarrow X)$. USAL thus trains its autoencoder by trying to find mapping functions ϕ, and ψ that minimize the reconstruction error L_{recon} such that,

$$\phi, \psi = \text{argmin}_{\phi,\psi}(\mathcal{L}_{recon}) \tag{2}$$

where,

$$\mathcal{L}_{recon} = \sqrt{\sum_{c}^{C}\sum_{t}^{T}(\mathbf{x}^{c,t} - \hat{x}^{c,t})^2} \tag{3}$$

In eq. (3) let $\mathbf{x}^{c,t} = [x_1^{c,t}, x_2^{c,t}, \ldots, x_m^{c,t}]$ be a vector representing a set of m inputs (e.g., voltage, current, temperature) given to USAL at some time t in a charge-discharge cycle c. Let $\hat{x}^{c,t} = [\hat{x}_1^{c,t}, \hat{x}_2^{c,t}, \ldots, \hat{x}_m^{c,t}]$ be a vector representing reconstructed inputs generated by the autoencoder at its output at the same time t in a charge-discharge cycle c. Thus, eq. 3 forms the first part of USAL's loss function, whose objective is to train an efficient autoencoder within USAL.

For under-complete autoencoders $\mathbf{h} \in \mathbb{R}^p$ where $p < m$ [27]. Typically, for the task of SOC estimations, current, voltage, and temperature are the only measured inputs. Thus, by choosing $p < m$, we may introduce a strong bias within USAL, limiting its SOC estimation capabilities. To overcome this problem, in this work, we use over-complete autoencoders where, $p \geq m$ [27], [28]. As per recommendations in literature [27], [30], [32], [33], to avoid learning identity functions, we enforce sparsity regularization over the encoding layer in the over-complete autoencoder within USAL. Identity function mapping can be avoided by limiting neuron activations in the encoding layer using L_1 regularization [34], [35]. Such sparsity-enforced over-complete autoencoders are known as 'Sparse Autoencoders' [33], [36]. In USAL, sparsity constraints are enforced using L_{spars} such that,

$$L_{spars} = \sum_{c}^{C}\sum_{t}^{T}|\mathbf{h}^{c,t}| \tag{4}$$

Here, $\mathbf{h}^{c,t} = [h_1^{c,t}, h_2^{c,t}, \ldots, h_p^{c,t}]$ represents the output of the encoding layer in USAL where, $h_p^{c,t}$ is the output of encoding p given some input $\mathbf{x}^{c,t}$.

B. Correlation Analysis

While L_1 regularization performs sparsity regularization, it offers little help with multicollinearity [37]. Multicollinearity is a phenomenon in which two or more variables are highly correlated [38]. Usage of such highly correlated encodings may severely limit the ability to extract aging information from the encodings [39]. In USAL, to ensure that the encodings learn a valuable representation of the input, we introduce the correlation loss $\mathcal{L}_{corrl}$, such that,

$$\mathcal{L}_{corrl} = \sum_{i}\sum_{j}|k_{i,j}| \tag{5}$$

where,

$$k_{i,j} = \frac{1}{C}\sum_{c}^{C}\frac{\sum_t^T (h_i^{c,t} - \bar{h}_i^c)(h_j^{c,t} - \bar{h}_j^c)}{\sqrt{\sum_t^T \left(h_i^{c,t} - \bar{h}_i^c\right)^2 \sum_t^T \left(h_j^{c,t} - \bar{h}_j^c\right)^2}} \tag{6}$$

In eq. 6, $k_{i,j}$ is the correlation between some encoding i and encoding j, and $\bar{h}_i^c$ is the mean output of some encoding i in cycle c. Note that $k_{i,j}$ is always equal to 1 if $i = j$. The correlation loss thus forms a crucial part of USAL and helps reduce multicollinearity within the encodings while simultaneously learning to transform inputs into outputs with minimal distortion.

C. Prediction Network

A prediction network with an LSTM layer forms the final part of USAL. LSTM networks are a variant of Recurrent Neural Networks (RNNs) that can learn temporal trends in the input data by developing and retaining memories from previous time steps [24], [26], [40], [41], [42], [43], [44]. USAL thus learns to perform accurate SOC estimations by

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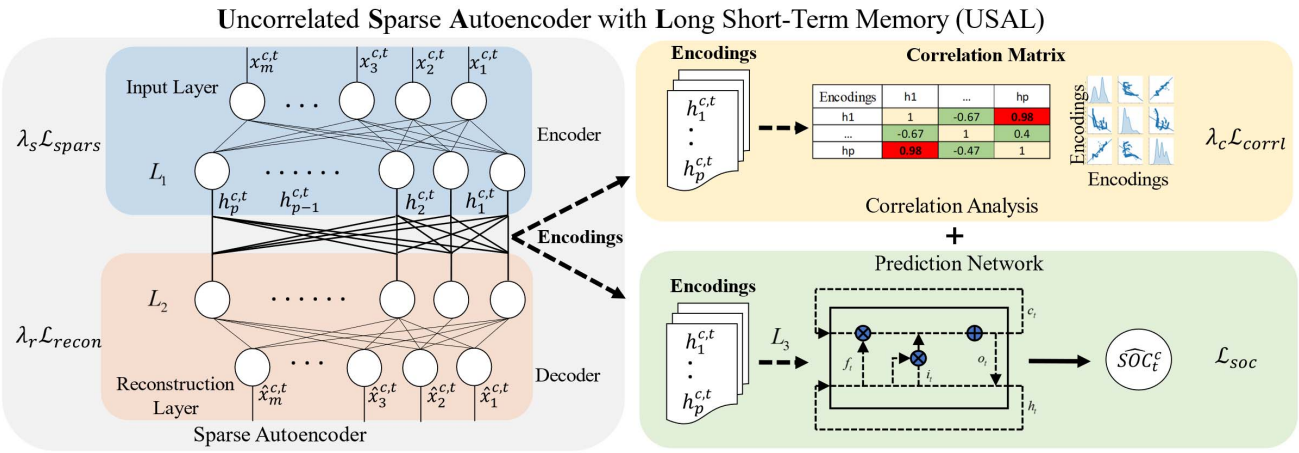


Fig. 3. Uncorrelated sparse autoencoder with long short-term memory (USAL) for state-of-charge estimations.

TABLE I
SUMMARY OF ACCELERATED AGING DATASETS

Dataset	Purpose	Cell Type (Cathode/Anode)	Nominal Capacity (Ah)	Nominal Voltage (V)
Dataset-1	Accelerated Aging	18650 LiB (LiCoO2/LiC)	2.0	4.2
Dataset-2		APR18650M1A (LFP/Graphite)	1.1	3.3
Dataset-3		Pouch Cell (LiCoO2/Graphite)	3.36	3.82

minimizing L_{soc}, such that,

$$\mathcal{L}_{soc} = \sqrt{\sum_{c}^{C}\sum_{t}^{T} \frac{\left(\widehat{SOC}_t^c - SOC_t^c\right)^2}{CT}} \tag{7}$$

In eq. 7, $S\hat{O}C_t^c$ is the SOC estimated by USAL and SOC_t^c is the true SOC given some input $\mathbf{x}^{c,t}$ in some cycle c at a time t.

In summary, given input $\mathbf{x}^{c,t}$ USAL simultaneously learns to recreate $\hat{x}^{c,t}$ using a sparse autoencoder and perform accurate SOC estimations ($S\hat{O}C_t^c$) using a prediction network while penalizing for high multicollinearity between the learned encodings. Equation 8 effectively summarizes the loss function used for training USAL where λ_r, λ_s and λ_c are the reconstruction, sparsity, and correlation regularization hyperparameters. Figure 3 helps visualize all components and their functions within USAL.

$$\mathcal{L}_{USAL} = \lambda_r \mathcal{L}_{recon} + \lambda_s \mathcal{L}_{spars} + \lambda_c \mathcal{L}_{corrl} + \mathcal{L}_{soc} \tag{8}$$

III. CASE STUDY

A. Datasets

This paper uses three publicly available accelerated aging datasets that have a wide range of applications [11], [45], [46], [47]. Every considered dataset follows a unique accelerated aging process and uses a unique battery configuration. The three considered datasets are heavily studied in the literature. Details of each dataset are summarized in Table I.

1) Dataset-1: The first dataset is provided by the NASA Ames Laboratory [45]. It collects batches of four Li-ion battery cells that are subject to accelerated aging using three different charge-discharge profiles. The recorded data includes measurements such as the current (in Amps), the voltage (in Volts), and the temperature (in °C) of each cell. For this study, from this dataset, we use cell 'B0005' for validation and hyper-parameter tuning and 'B0007' for testing. For all Li-ion battery cells in this dataset, charging is carried out in Constant Current (CC) mode at 1.5A until the cell voltage reaches 4.2V. It is then continued in a Constant Voltage (CV) mode until the charge current drops to 20mA. The battery cells are then discharged at a Constant Current (CC) level of 2A until the cell voltage drops to 2.7V. Finally, repeated charging and discharging is implemented, which results in the accelerated aging of the battery cells. Experiments are stopped when the battery cells reach EOL- 30% fade in rated capacity. This dataset is referred to as 'Dataset-1'.

2) Dataset-2: The second dataset provided by Severson et al. [46] is a congregation of 124 LFP/Graphite battery cells (A123 Systems, model APR18650M1A). The recorded data includes measurements such as current (in Amps), voltage (in Volts), and temperature (in °C) for each battery cell. This dataset aims to explore the effects of fast charging on cell aging. The battery cells are charged from 0% to 80% SOC with one of 72 different one-step and two-step charging policies. They are then charged from 80% to 100% SOC with a uniform 1C CC-CV charging step to 3.6V and a current cut-off of C/50 where 1C is 1.1A. During the discharge process, the battery cells are discharged under the CC-CV conditions with discharge current set to 4C and 2.0V with a current cut-off of C/50. Repeated charging and discharging results in the accelerated aging of the battery cells. Experiments are stopped when the battery cells reach EOL- 20% fade in the rated capacity. This dataset is referred to as 'Dataset-2'.

3) Dataset-3: The third dataset provided by Diao et al. [47] is a collection of 192 Li-ion battery cells. In their experiments, Diao et al. [47] perform accelerated aging of the battery cells using three stress factors and a full factorial

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design of experiments framework. The recorded data includes measurements such as the current (in Amps) and each battery cell's voltage (in Volts). This dataset aims to analyze the effect of these stress factors on capacity fade by continuously cycling cells. The three stress factors used are- (i) ambient temperature (10 °C, 25 °C, 45 °C, 60 °C), (ii) charge cut-off C-rate (C/5, C/40), and (iii) discharge current rate (0.7C, 1C, 2C). In their experiments, the battery cells are charged at 1.5C to 4.2V. They are then held at 4.2V until the current decreases to 1C (3.36A). Charging is then continued up to 4.4V at 1C. The battery cells are held at 4.4V until the current drops to the given charge cut-off rate in stress factor (ii). The battery cells are rested for five minutes and finally discharged using the discharge rate given by the stress factor (iii). Repeated charging and discharging is implemented, which results in the accelerated aging of the battery cells. Experiments are stopped when the battery cells reach EOL- 50% fade in the rated capacity. Note that the tests are performed at ambient temperature values given by stress factor (i). This dataset is referred to as 'Dataset-3'.

The datasets used in this study represent a comprehensive coverage of experimentation in accelerated aging for Li-ion cells. Figure 4 shows the charge-discharge profiles of all test Li-ion battery cells used in this study. More details on the specific test conditions of every considered Li-ion battery cell are provided and discussed in the following sections.

B. Training and Tuning USAL

For long-term SOC estimations, USAL is trained using the initial five and tested on the final twenty charge-discharge cycles for every considered Li-ion battery cell, unless specified otherwise. Table II summarizes the inputs to USAL along with details of training and testing data for all three accelerated aging datasets used in this paper. For Li-ion battery cells in 'Dataset-2', we introduce the moving average of the last five time steps for the measured current and voltage as two additional inputs based on the recommendations of Chemali et al. [6]. Figure 5 can be used to visualize the flow of information within USAL along with its loss components for a typical Li-ion battery cell.

USAL is a highly customizable deep neural network that can be modified according to the task on hand. In this section, this task is long-term SOC estimations. We use data from cell 'Cell-B0005' in 'Dataset-1' for tuning USAL's architecture and hyper-parameters. Although several architectures for USAL were tried in this case study, we only provide details for the best-performing architecture. All tested models were trained using the same initializations for a fair evaluation. The sparse autoencoder within USAL is an LSTM encoder-decoder network that consists of two key LSTM layers L_1 and L_2. In our experiments, although several values of p were tried, p, when set to 10, gave the best results unless specified otherwise. Thus, both L_1 and L_2 are defined as LSTM layers with 10 nodes each. Further, the prediction network within USAL consists of a single LSTM layer, L_3 with 10 nodes and uses the encodings generated by the encoder within USAL as inputs. The lookback steps for all LSTM layers within USAL are set to 50. While the outputs of all layers in USAL are transformed using a 'tangent hyperbolic' function, the outputs of the reconstruction layer are transformed using a 'linear' activation function.

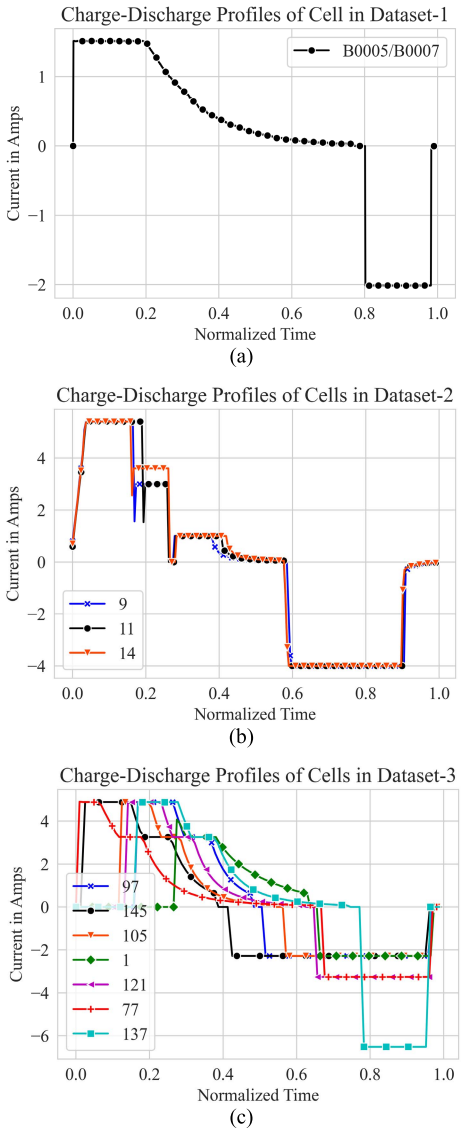


Fig. 4. Charge and discharge profiles of all cells used in this study in – (a) Dataset-1, (b) Dataset-2, and (c) Dataset-3.

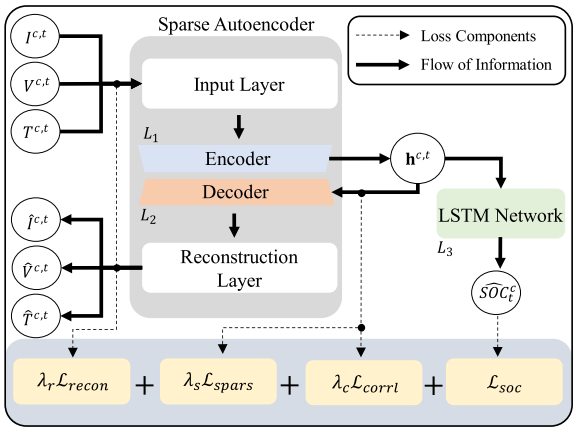


Fig. 5. Flow of information within USAL and its loss components using commonly measured inputs in a typical Li-ion battery cell.

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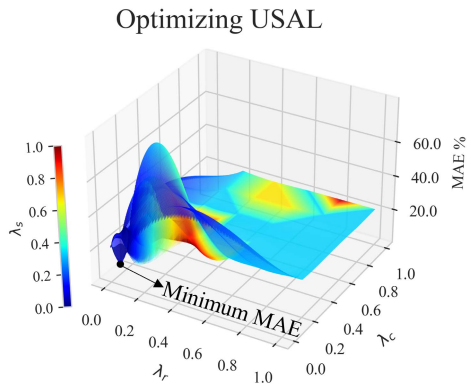


Fig. 6. Long-term SOC estimation validation performance using different values of λ_r, λ_s and λ_c.

TABLE II
SUMMARY OF TRAIN-TEST DATA

Dataset	Train Cycles	Test Cycles	Inputs to USAL
Dataset-1	Initial 5 Cycles	Final 20 Cycles	$I^{c,t}, V^{c,t}, T^{c,t}$
Dataset-2			$I^{c,t}, V^{c,t}, T^{c,t}, I^{c,t}_{avg}, T^{c,t}_{avg}$
Dataset-3			$I^{c,t}, V^{c,t}$

As in any deep neural network, hyper-parameter tuning is an essential aspect of USAL. As such, in our experiments, we run an exhaustive grid search and train-test USAL by varying λ_r, λ_s and λ_c from 0-1 in eq.8. Figure 6 can be used to visualize the performance of USAL for long-term SOC estimations when using validation 'Cell-B0005'. As can be seen in this figure, USAL shows its best performance when λ_r, λ_s and λ_c are set to 0.06, 1e-4, and 1e-3, respectively. As can be observed in Fig. 6, USAL shows poor performance when either λ_r, λ_s and λ_c are set to zero. This provides evidence that all losses defined in Section II play a crucial role in training USAL. Thus, we use the above-defined architecture, hyper-parameter values, and training strategy in all our experiments when performing SOC estimations using USAL. Note that we train USAL for 300 epochs in all our experiments using the 'Adam' optimizer on an Intel Xeon 5218R CPU and a single Nvidia RTX 2080 Ti GPU.

C. Comparison Across Battery Configurations

Proposition of models that estimate SOC only for specific battery cell configuration offers limited value to the industry. While one manufacturer may prefer a LiCoO2 cathode, the others may prefer an LFP cathode. To this end, we train and test USAL on three Li-ion battery cells with varying configurations. Specifically, we use 'Cell-B0007' from 'Dataset-1', 'Cell-11' from 'Dataset-2', and 'Cell-77' from 'Dataset-3'. We compare the performance of USAL with three other data-driven models- a Deep Neural Network (DNN), a standard LSTM network, and another autoencoder-based SOC estimation framework, the Autoencod-LSTM [26]. Although several architectures for DNN and LSTM networks were tried,

TABLE III
SUMMARY OF RESULTS UNDER VARYING CELL CONFIGURATIONS

Ambient Temperature (°C)	Cathode/Anode	Cell Id	MAE (%)	MAX Abs. Error (%)	RMSE (%)
Room Temp.	LiCoO2/LiC	B0007	3.63	69.10	6.63
25	LFP/Graphite	11	3.08	16.12	4.06
25	LiCoO2/Graphite	77	0.88	3.80	1.10

we provide details only for the best-performing architectures. The DNN modeled herein consists of three hidden layers with 16, 8, and 4 neurons each. The standard LSTM network used for comparisons consists of a single LSTM layer with 20 hidden nodes with 50 lookback time steps. The DNN and the standard LSTM network are trained using the mean squared error loss function for 300 epochs. We use the same architecture and strategy to train the Autoencod-LSTM as given by Fasahat et al. [26]. For a fair comparison, all benchmarked models use the same data used to train and test USAL.

Figure 7(a) compares the performance of USAL with the three data-driven models for long-term SOC estimations for battery cells given in Table III. Long-term SOC estimations near the EOL are most challenging, given only a few charge-discharge cycles. Nevertheless, as shown in Fig. 7(a), USAL comfortably exceeds the other data-driven models' performance. The poor performance of DNNs can be attributed to their lack of ability to capture temporal trends in the data. While the standard LSTM network and Autoencod-LSTM improve on this, they cannot learn the degradation and aging information from just five training cycles. Furthermore, due to a two-step training procedure used in the Autoencod-LSTM, there is no guarantee that the generated encodings will be tailored specifically for long-term SOC estimations. However, USAL applies a multi-task training procedure that penalizes simultaneously for (i) poor SOC estimations, (ii) poor reconstructions, and (iii) highly correlated encodings for accurate long-term SOC estimations. As a result, USAL outperforms other benchmarked methods. For example, while the Autoencod-LSTM achieves an MAE of 2.1%, the USAL achieves a significantly lower MAE of 0.88% for 'Cell-77'. Figure 8(a) can be used to visualize this difference in the performance for 'Cell-11.' Using this figure, we can also observe that estimations produced by USAL are less noisy than the other data-driven models. Hence, we can conclude that battery cell configurations have no adverse effect on USAL's ability to perform long-term SOC estimations. Table III summarizes the performance of USAL using varying cell configurations.

D. Comparisons Across Varying Charge, Discharge, and Temperature Conditions

Devices such as autonomous vehicles may frequently charge and discharge their battery cells under varying operational conditions. It is well-known that Li-ion battery cells age differently under varying charging, discharging, and ambient temperature conditions [23], [47]. Therefore, in this

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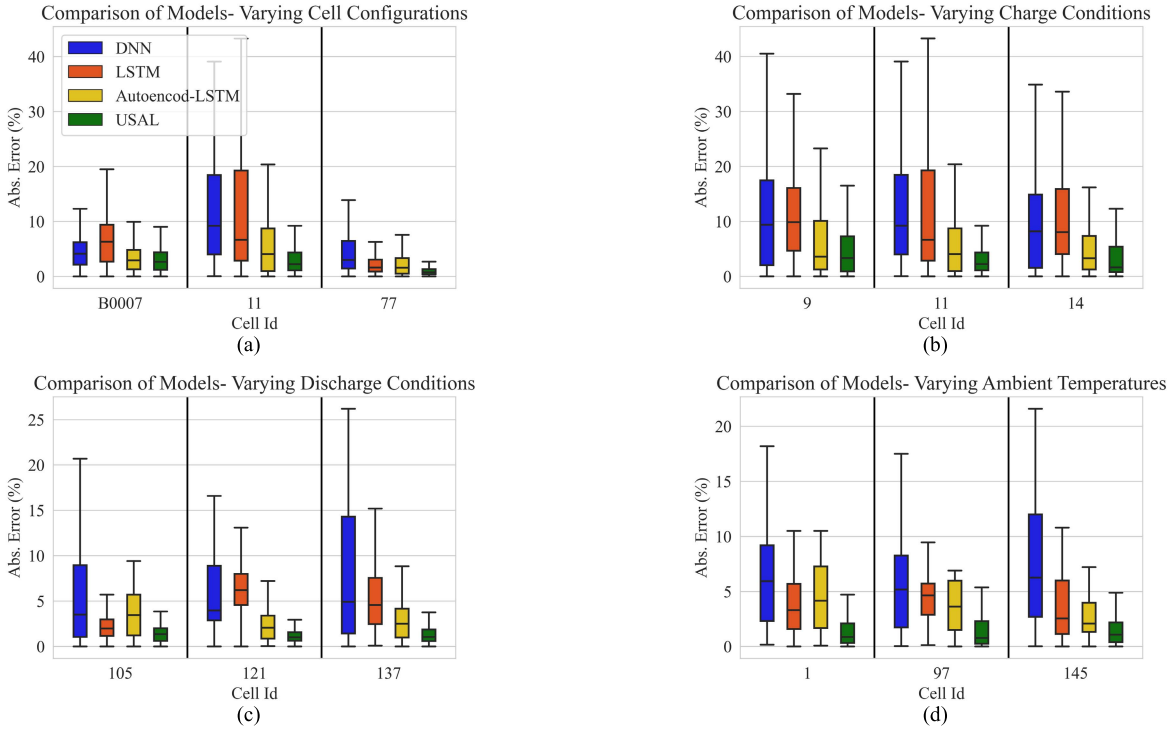


Fig. 7. Comparison of USAL with benchmarked models across varying: (a) Cell configurations, (b) Charge conditions, (c) Discharge conditions, and (d) Ambient temperatures across twenty test cycles near EOL.

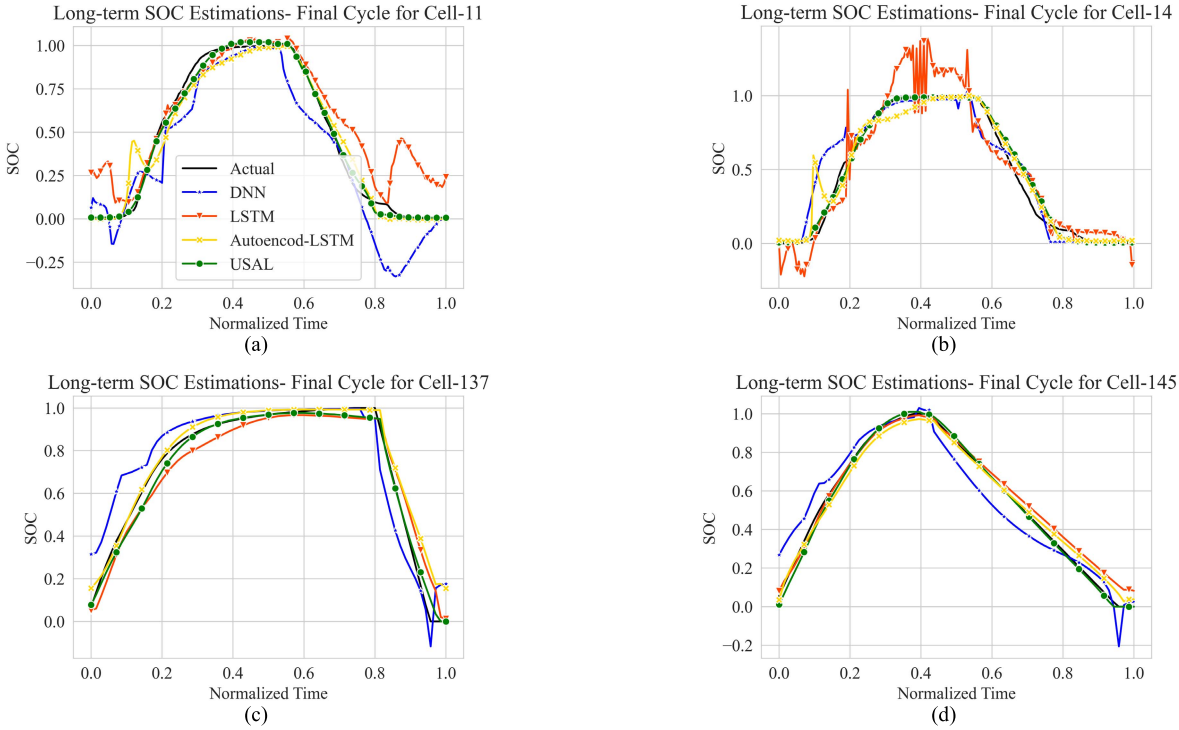


Fig. 8. Long-term SOC estimations for final cycle near EOL for (a) Cell-11, (b) Cell-14, (c) Cell-137 and (d) Cell-145.

section, we perform long-term SOC estimations on battery cells aged under various operating conditions. Battery cells and their respective operational conditions are given in Table IV.

Figures 7(b) and 7(c) compare the performance of USAL with the other data-driven models developed in Section III.C for long-term SOC estimations. From these figures, it can be observed that while the other data-driven models are sensitive

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TABLE IV
CELLS AND THEIR OPERATIONAL CONDITIONS

Dataset	Cell Id	Charge	Discharge	Ambient Temperature
Dataset-2	9	5.4C(40%)-3C	4 C	30°C
	11	5.4C(50%)-3C		
	14	5.4C(60%)-3.6C		
Dataset-3	105	C/40	0.7 C	45°C
	121		1 C	
	137		2 C	
	1	C/5	0.7 C	10°C
	97			45°C
	145			60°C
	77	C/40	1 C	25°C

TABLE V
SUMMARY OF RESULTS UNDER VARYING CONDITIONS

Dataset	Cell Id	MAE (%)	MAX Abs. Error (%)	RMSE (%)
Varying Charge Conditions				
Dataset-2	9	4.55	30.91	6.22
	11	3.08	16.12	4.06
	14	3.51	17.49	5.12
Varying Discharge Conditions				
Dataset-3	105	1.43	5.29	1.74
	121	1.29	4.48	1.66
	137	1.67	16.71	2.54
Varying Ambient Temperatures				
Dataset-3	1	1.38	4.71	1.90
	97	1.51	5.93	1.22
	145	1.57	5.78	2.24

to changes in the charge-discharge patterns, USAL and its estimations are robust to any such changes. Extreme charge and discharge conditions age Li-ion battery cells quite quickly, leading to a steep drop in their capacity with time. As a result, it is hard to estimate SOC near EOL. This can be observed in the long-term SOC estimations for 'Cell-14' and Cell-137, which are showcased in Fig. 8(b) and Fig. 8(c), respectively. While other data-driven methods are unlikely to perform well in such situations, the encodings produced by USAL help overcome this drawback. This can be observed in estimations for 'Cell-14' in Fig. 8(b). Thus, using results from Fig. 7 and Fig. 8, we can observe that USAL demonstrates sufficient ability to capture the Li-ion battery cell's aging trends for long-term SOC estimations under varying charging and discharging conditions.

Li-ion battery cells are known to age quite significantly under higher ambient temperatures [21], [46], [47]. Therefore, it is of our high interest to validate the long-term SOC estimation performance of USAL aged under the same charge-discharge profiles but at different ambient temperatures. In our experiments across three Li-ion battery cells, a familiar pattern emerges, and USAL outperforms the other data-driven models for long-term SOC estimations under varying ambient temperatures. Out of three tested battery cells, 'Cell-145' ages quickly due to a higher ambient temperature (60 °C). Thus, as shown in Fig. 7(d) and Fig. 8(d), it is much harder for the benchmarked data-driven models to perform accurate long-term SOC estimations. On the other hand, USAL shows overwhelming accuracy in its estimations, achieving a maximum absolute error of 5.78% in its SOC estimations over the twenty test cycles for 'Cell-145'. Table V summarizes the performance of USAL across varying charging, discharging, and ambient temperature conditions.

E. Short-Term SOC Estimations

The SOC estimation module is responsible for long-term and short-term SOC estimations within the BMS. Therefore, a SOC estimation model that performs excellent long-term SOC estimations while suffering from poor performance in short-term SOC estimations is of little practical value. Thus, in this section, we perform short-term SOC estimations using 'Cell-137' in 'Dataset-3'. Here, all models are tested on the 6th charge-discharge cycle. Figure 9 can be used to visualize the performance of all models for short-term SOC estimations. In Fig. 9(a), we can observe that the standard LSTM shows acceptable performance while estimating the 6th cycle in 'Cell-137' with an MAE of 1.05%. The Autoencod-LSTM improves over this and achieves an MAE of 0.79%. The DNN cannot perform such accurate estimations due to its lack of ability to capture time-based trends. As seen in Fig. 9(a), USAL improves upon the performance of all benchmarked models achieving an MAE of 0.65% for short-term SOC estimations. The higher dimensional encodings and USAL's ability to negotiate multicollinearity within these encodings result in noise-free and accurate short-term estimations. Figure 9(b) showcases the estimation errors for all models for 'Cell-137.' As the disconnect between the training and test cycles increases, the performance of the benchmarked data-driven models worsens. USAL, on the other hand, is much more accurate and reliable. For example, while the Autencod-LSTM achieves an MAE of 6.1% on the last cycle near EOL for 'Cell-137,' USAL estimates this cycle much better, achieving an MAE of 2.5%. The SOC estimations near EOL and associated estimation errors for 'Cell-137' can be seen in Fig. 9(c) and Fig. 9(d), respectively.

F. Effects of Correlations on Long-Term SOC Estimations

In our experiments, although USAL is shown to exceed performance over other benchmarked models, it is of our high interest to validate that this performance gain is due to the reduced multicollinearity between the learned encodings. Consequently, we choose our most challenging cells – 'Cell-14', 'Cell-137' and 'Cell-145' given in Table V and compare the performance of USAL with and without the correlation loss. We specifically choose these three cells since they showcase the most rapid drop in capacity due to extreme test conditions. The models are then trained using the same architecture and training strategy as given in Section III.B, and correlation loss is disabled by setting λ_c to zero in eq.8. Note that we use the same initializations when training with and without correlation loss for a fair comparison.

As can be observed in Table VI, without the correlation loss, the performance of USAL worsens across all tested battery

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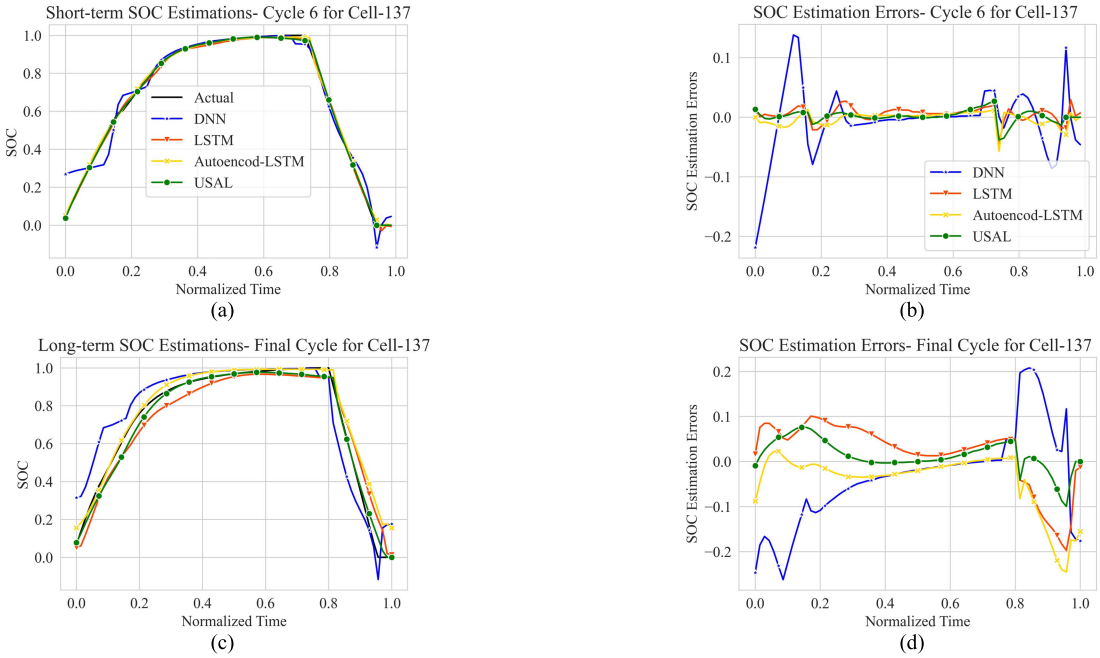


Fig. 9. Comparisons for short-term and long-term SOC estimations in 'Cell-137' for USAL and other benchmarked models.

TABLE VI
EFFECTIVENESS OF CORRELATION LOSS

Cell Id	MAE (%)	MAX Abs. Error (%)	RMSE (%)	Mean Abs. Correlation Between Encodings
Without Correlation Loss				
14	3.75	19.39	5.43	0.65
137	7.70	19.54	8.17	0.52
145	4.13	12.98	5.17	0.59
With Correlation Loss				
14	3.51	17.49	5.12	0.32
137	1.67	16.71	2.54	0.34
145	1.57	5.78	2.24	0.34

cells. As the correlation between the encodings reduces, the sparse autoencoder within USAL learns better representations of input data. Such unique data representations can then be leveraged to simultaneously reconstruct the inputs while identifying long-term aging effects for accurate SOC estimations. Learning such informative, higher-dimensional encodings is sensitive to the correlation between encodings. While the correlation loss increases the diversity between the learned encodings, the sparsity, reconstruction, and prediction losses force the network to learn significant aging trends when presented with limited training data. Results in Table VI reinforce this intuition.

G. Robustness Evaluation

Typical, off-the-shelf measurement devices often exhibit noise levels ranging from 2% to 4% in their measurements [6]. Thus, any SOC estimation method must be robust against such noise in the measured inputs. As a result, we evaluate the robustness of USAL using the same approach as Chemali et al. [6]. In particular, we inject random Gaussian noise with zero mean and a standard deviation of 4% into the measured inputs. When presented with such noisy inputs, USAL achieves its worst performance with an MAE of 3.88% for 'Cell-14'. On the other hand, the Autoencod-LSTM achieves an MAE of 5.4% when no noise is injected in measured inputs for 'Cell-14.' A similar pattern is also observed for 'Cell-137' and 'Cell-145', where the MAE achieved by USAL with injected noise (3.87% and 2.55%) are lower than MAE achieved by Autoencod-LSTM with no injected noise (4.37% and 2.75%). We can thus conclude that USAL is indeed a robust method that performs accurate long-term SOC estimations even under the presence of noisy inputs.

TABLE VII
ROBUSTNESS OF USAL

Cell Id	MAE (%)	MAE (%) with noise	RMSE (%)	RMSE (%) with noise
14	3.51	3.88	5.12	5.50
137	1.67	3.87	2.54	4.93
145	1.57	2.55	2.24	3.40

H. SOC Estimations for Dynamic Loading Profiles

State-of-art SOC estimation models are often benchmarked on dynamic loading profiles like Dynamic Stress Test (DST), Federal Urban Driving Schedule (FUDS), US06 Highway Driving Schedule (US06), and Beijing Dynamic Stress Test (BJDST). This helps prove the model's estimation capabilities in real-life loading conditions. However, high input variability combined with random charging and discharging patterns make SOC estimation challenging for dynamic loading profiles. Although this work aims to develop a long-term SOC

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TABLE VIII
COMPARISON OF USAL WITH STATE-OF-THE-ART

Method	Train Data	Test Data at 25°C	RMSE (%)	MAE (%)
GPR [48]	FUDS- First 3000 samples at 25°C	US06- First 3000 samples	0.76	2.25
Recurrent GPR w/ two-tap delay [48]			0.63	0.82
Autoregressive Recurrent GPR [48]			0.24	0.81
USAL			**0.59**	**0.50**
Stacked LSTM [43]	DST, US06 at 25°C	FUDS	1.07	0.84
Autoencod-LSTM [26]			1.65	0.99
USAL			**0.60**	**0.48**
Stacked LSTM [43]	FUDS, US06 at 25°C	DST	1.39	2.02
Autoencod-LSTM [26]			1.00	0.66
USAL			**0.57**	**0.45**
SSMPI [49]	DST at 0, 25, and 45°C	US06	-	0.94
USAL			**0.99**	**0.81**
SSMPI [49]		BJDST	-	0.66
USAL			**0.66**	**0.53**

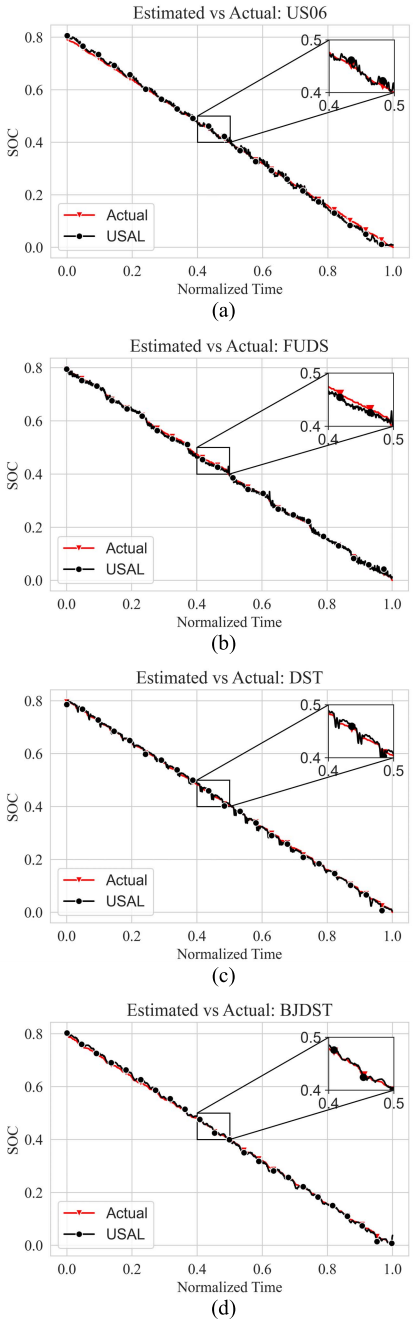


Fig. 10. SOC estimation performance of USAL for – (a) the US06 cycle, (b) the FUDS cycle, (c) the DST cycle, and (d) the BJDST cycle.

estimation model, evaluating its performance on dynamic loading profiles is crucial. To this effect, we use a fourth public dataset provided by the Center for Advanced Life Cycle Engineering (CALCE) at the University of Maryland [11]. This dataset uses the 'INR 18650-20R' cells (LiNiMnCo/Graphite) with a capacity rating of 2Ah. Initially, the cells are charged and then discharged so that the SOC reaches 80%. The cells are then loaded based on the selected loading profile. Current (in Amps) and voltage (in Volts) are logged every second in this test. The true SOC is calculated every second using the 'Coulomb Counting' method. Further, the architecture of USAL is optimized based on the task at hand, which in this case is SOC estimations under dynamic loading conditions. Although several architectures for USAL were tried, we only provide details for the best-performing architecture in this case study. For hyper-parameter tuning, USAL was trained using data from the DST cycle recorded at 45°C and validated using data from the BJDST driving cycle recorded at 25°C. USAL achieved its best performance when p was set to 20. Hence both L_1 and L_2 are defined as LSTM layers with 20 nodes each. Further, the prediction network within USAL was reconfigured to have an LSTM layer (L_3) with 20 nodes. For all layers within USAL, the lookback steps are set to 50. USAL achieved its best results when no changes were made to output functions of neural network layers, λ_r, λ_s and λ_c. These values are thus set to the same values as given in Section III.B.

Table VIII can be used to compare the performance of USAL with other state-of-art methods [43], [48], [49]. USAL was trained and tested on the same data as in Table VIII for a fair comparison with other benchmarked methods. Using this table, we can observe that USAL outperforms all other benchmarked methods in all our experiments. Figure 10 can be used to visualize this SOC estimation performance of USAL for the US06, FUDS, DST, and the BJDST driving cycle. These performance gains can be directly attributed to USAL's ability to learn implicit, uncorrelated trends in input data. Autoencod-LSTM [24] is a step-by-step method, while the proposed method (USAL) is integrated where each part of the model is trained simultaneously. A comparison of Autoencod-LSTM with USAL (and correlation loss) in Table IX helps illustrate the contributions of this work in real-world applications. A comparison of Autoencod-LSTM [26] with USAL but without correlation loss ($\boldsymbol{\lambda_c = 0}$) in Table IX, helps illustrate the effectiveness of the proposed multi-task training strategy. Thus, we can conclude that USAL is a state-of-art method supporting SOC estimations even under dynamic driving conditions. Furthermore, USAL's size (<100KB) and its ability to capture aging and degradation trends make it a state-of-art algorithm that can be integrated

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TABLE IX
EFFECTIVENESS OF USAL

Method	Train Data	Test Data at 25℃	RMSE (%)	MAE (%)
Autoencod-LSTM [26]	DST, US06 at 25℃	FUDS	1.65	0.99
USAL ($\boldsymbol{\lambda_c = 0}$)			0.68	0.56
USAL			**0.60**	**0.48**
Autoencod-LSTM [26]	FUDS, US06 at 25℃	DST	1.00	0.66
USAL ($\boldsymbol{\lambda_c = 0}$)			0.66	0.54
USAL			**0.57**	**0.45**

into a modern BMS for accurate and reliable SOC estimations. In our experiments, the average training time for USAL was about 1200 secs (300 epochs), and the average estimation time was about 0.65ms per observation.

IV. CONCLUSION

This paper introduces the Uncorrelated Sparse Autoencoder with Long Short-Term Memory (USAL) for long-term SOC estimations in Li-ion battery cells. Unlike existing methods [6], [24], [26], the use of a multi-task training strategy in USAL helps it learn efficient higher-dimensional encodings given limited training data. Such encodings may then be used to identify the effect of temporal and aging patterns for accurate SOC estimations. While existing methods fail to identify cell aging and degradation-related changes, the correlation, reconstruction, sparsity, and prediction penalties within USAL force diversity in the learned encodings by penalizing for high multicollinearity. Based on our experiments, USAL routinely outperforms other data-driven models for long-term SOC estimations. Significant improvements are observed when evaluated under varying datasets, cell configurations, charge-discharge profiles, and ambient temperature conditions. USAL is also shown to outperform benchmarked state-of-the-art methods when tested using dynamic loading profiles.

While the proposed method seems to have a high implementation cost, this cost is only a one-time cost. In addition, the benefits that can be achieved by implementing this method in a BMS are priceless, especially with rise propositions such as 'cloud-based' BMS systems [48], [50]. For future work, variations of USAL for Remaining Useful Life (RUL) and battery reliability predictions are of high interest [5], [16], [51].

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